



University of Vavuniya

Second Examination in Information Communication Technology - 2020

First Semester - May/June 2022

TICT2113 - Data Structures and Algorithms (Theory)

Answer Four Questions Only

Time Allowed: Two hours

1. (a) Briefly describe what are *linear* and *non-linear* data structures with suitable examples. [10%]
- (b) Briefly describe the strategy “*divide and conquer*” and state a sorting algorithm in which the divide and conquer strategy is used. [10%]
- (c) Perform the *Insertion sorting* algorithm to trace the following unsorted list:
- | | | | | | | |
|----|----|----|----|----|---|----|
| 78 | 34 | 45 | 18 | 28 | 6 | 69 |
|----|----|----|----|----|---|----|
- [30%]
- (d) Write a Java code for each of the following methods to search an element from an array of integers:
- Linear search.
 - Binary search.
- [30%]
- (e) Apply Linear search and Binary search to search an element 45 from the sorted list given below:

6	18	28	34	45	69	78
---	----	----	----	----	----	----

[20%]

2. (a) State the main difference between a *tree* and a *binary tree* in the context of data structures. [10%]

- (b) Consider the following binary tree in Figure 1:

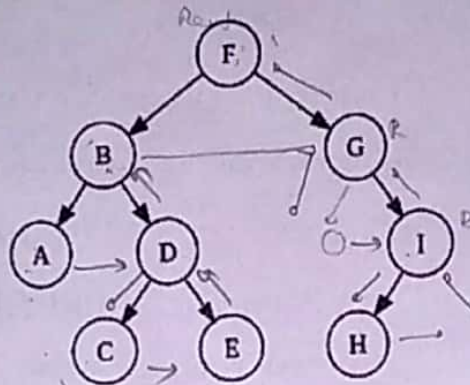


Figure 1:

Write down the order of visited nodes when the tree is traversed by each of the following methods:

- Pre-order method. $\rightarrow F \rightarrow B \rightarrow A \rightarrow D \rightarrow C \rightarrow E \rightarrow G \rightarrow I \rightarrow H$
- In-order method. $\rightarrow A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow H \rightarrow I$
- Post-order method. $\rightarrow A \rightarrow C \rightarrow E \rightarrow D \rightarrow B \rightarrow H \rightarrow I \rightarrow G \rightarrow F$

[30%]

- (c) Represent the binary tree shown in Figure 1 into each of the following methods:

- List Representation. $(F(B(A(D(C(E)))G(I(H))))$
- Left Child - Right Sibling Representation.

[30%]

- (d) Insert the following sequence of keys in the hash table using the hash function 'key mod 7'.

50, 700, 76, 85, 92, 73 and 101

Use linear probing technique for collision resolution.

[30%]

Array
Index
1 1 0 1 6 1 1 3 3

50/7=7R1
700/7=100R0
76/7=10R6
85/7=12R1
92/7=13R1
73/7=10R3
101/7=14R3

✓ 3. (a) Define each of the following terms:

i. Complete Binary Tree

ii. Binary Search Tree

iii. Full Binary Tree

[15%]

(b) Draw a binary search tree (BST) for a given data below:

108	142	8	97	39	15	44	126	21	58	62
-----	-----	---	----	----	----	----	-----	----	----	----

[20%]

(c) Perform the following tasks in the resultant binary tree generated from part(b).

i. Traverse the above BST by using all three methods and list out the resultant sequences.

[15%]

ii. Insert a new node 32 in the above BST.

[10%]

iii. Delete a node 21 from the above BST.

[10%]

(d) Differentiate **max heap** and **min heap** tree data structures with aid of examples.

[10%]

* (e) Consider the following max heap tree, and perform the following tasks given below.

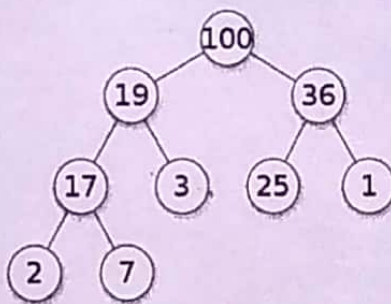


Figure 1:

i. Insert 43 and 21 into the tree and clearly trace the steps.

[10%]

ii. Delete the maximum element from the above max heap tree structure.

[10%]

In

8 → 15 → 21 → 39 → 44 → 58 → 62 → 97 → 108 → 126 → 142

Pre

108 → 8 → 97 → 39 → 15 → 21 → 44 → 58 → 62 → 142 → 126

Post

21 → 15 → 62 → 58 → 44 → 39 → 97 → 8 → 126 → 142 → 108

4. (a) Compare and contrast **Stack** and **Queue** data structures. [20%]

(b) Describe the concept of **Queue** data structure with aid of a real world problem. [20%]

(c) Define and implement each of the following Abstract Data Type operations of a Queue data structure using an array:

i. *enQueue()* - Inserting an element to the Queue.

ii. *deQueue()* - Removing an element from the Queue. [30%]

(d) Suppose that, a stack S (integer type) and a queue Q (integer type) are used to keep integers and initially they are empty.

Graphically describe the status of S and Q after each of the following operations:

S.push(60)

S.push(18)

Q.enQueue(23)

Q.enQueue(36)

x = S.pop()

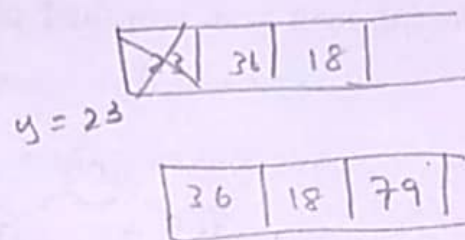
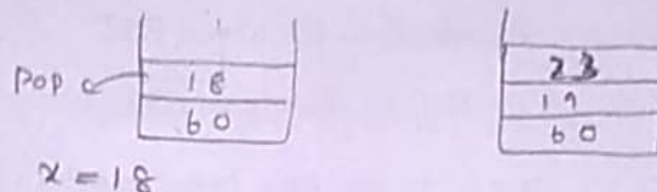
S.push(19)

Q.enQueue(x)

y = Q.deQueue()

Q.enQueue(79)

S.push(y)



[30%]

- ✓ 5 (a) Briefly describe what are *Singly linked list* and *Doubly linked list* data structures. [20%]

- (b) Consider the following singly linked list in Figure 1:

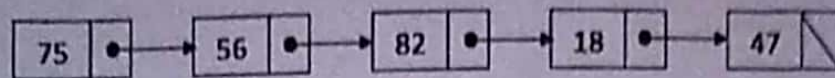


Figure 1:

Write the steps to perform the following operations based on the above singly-linked list:

- To insert a new node as a head node in the singly linked list.
 - To remove the middle element 82 from the singly linked list.
 - To insert a new node as a tail node in the singly linked list.
 - To make linear singly linked list to circular linked list. [40%]
- (c) Evaluate each of the following arithmetic expressions using *infix - postfix conversion* method in table format and show every status of stack in each of the following expressions.

i. $2 * 3 / (2 - 1) + 5 * 3.$ $2\ 3\ *\ 2\ 1\ -\ 5\ 3\ *\ +\ /$

ii. $(A + B * (C - D)) / E.$ $A\ B\ C\ D\ -\ *\ +\ E\ /\$ [40%]



University of Vavuniya

Second Examination in Information Communication Technology - 2020

First Semester - May/June 2022

TICT2113 - Data Structures and Algorithms (Practical)

Answer All Questions

Time Allowed: Two hours

✓ 1. (a) Write a Java class to implement the circular queue data structure using an integer array to perform each of the following operations.

- i. `isFull()` - Return true if the circular queue is full and return false otherwise.
- ii. `isEmpty()` - Return true if the circular queue is empty and return false otherwise.
- iii. `enqueue(x)` - Insert an element x into the circular queue.
- iv. `dequeue()` - Remove an element from the circular queue.
- v. `displayQueue()` - Display the circular queue elements from front to rear end.

[25%]

✓ (b) "The Call Center of a phone system uses Queues to hold the people calling them until a service representative attends the call."

Write a Java class with the above methods to perform each of the following tasks using the circular queue implemented in part (a) and assume that it maintains up to 5 callers in the queue.

[This question is continued on the next page]

i. Insert the following callers' ID into the circular queue in the given order.

2230 , 2840 , 3630 , 4540 , 5652

ii. Remove one caller ID from the circular queue and display it.

iii. Add new caller ID 2330 into the circular queue.

iv. Display the callers' ID available in the calling Queue.

[25%]

2. (a) Write a Java program to perform each of the following tasks.

↪ i. Sort a given array of elements using *Selection sort* technique.

✕ ii. Search an element using *recursive binary search* technique.

[30%]

↗ (b) Trace the recursive binary search algorithm to search 48 in the given array.

24	33	12	08	18	15	48	36	27
----	----	----	----	----	----	----	----	----

[20%]

2 < 9
1 < 9
0 < 9

max < arr[j]
24 < 33

while (n < arr.length)

val = arr[n]

Time: Two hours

✓ 1. Human-Computer Interaction (HCI) is critical since it will be necessary for goods to be more successful, safe, helpful, and functional.

- a) What is HCI? [15%]
- b) What is the difference between Graphical User Interface (GUI) and Command Line Interface (CLI)? [20%]
- c) Discuss five characteristics of the future of the HCI. [20%]
- d) Explain each of the following terms in the context of interaction design used in HCI. [20%]
 - i. domain
 - ii. goal
 - iii. task
- (e) "Human factors and ergonomics are the application of psychological and physiological principles to the engineering and design of products, processes, and systems". Discuss four factors pertaining to humans in the HCI process. [25%]

✓ 2. The basic goal of interaction design is to create usable interactive products.

- a) What do you mean by *usability*? [15%]
- b) State *three* principles to support usability. [15%]
- c) "Website usability is measured by using performance and perception attributes". Discuss performance attributes including effectiveness, efficiency, and errors and perception attribute including satisfaction. [30%]
- d) "Think global, act local" is a strategy to gain global market share. The researchers suggest website designers and developers accommodate cultural preferences in

websites, to succeed globally. As a website designer, how can you plan for the website's globalization and localization. [20%]

- e) Explain four challenges for the website's cultural adaptation, for the Asian market. [20%]

✓ 3.

- a) Compare and contrast: "*User Centered Design(UCD)*" and "*User experience design(UXD)*" in the context of HCI. [20%]
- b) Miller's law, proposed by George Miller, tells us that we can only keep around seven items in our working memory". Discuss how Miller's Law can help to enhance user experience. [20%]
- c) Discuss the use of Flat vs. Deep website hierarchies, with appropriate examples. [20%]
- d) Discuss the Norman's seven principles with the aid of examples. [20%]
- e) Discuss four emerging trends in mobile user interfaces. [20%]

4.

- a) Design rationale is information that explains why a computer system is the way it is. List down four benefits of design rationale. [15%]
- b) Rittel and Kunz's Issue-based Information System (IBIS) is an important approach to argumentation of design rationale. Discuss the graphical IBIS, with appropriate examples. [25%]
- c) Discuss the Design Space Analysis using suitable examples. [25%]
- d) Automated usability evaluation provides a number of advantages and overcomes the limitations of its non-automated counterpart. Explain how you can conduct an Automated Usability Evaluation to evaluate website usability. [35%]

5.

- ✓ a) List down five Universal design principles. [20%]
- b) Discuss three advantages and disadvantage with speech recognition technology. [20%]
- + c) Describe four conventional methods of making users notice messages when interacting with interfaces. [20%]
- d) Colour-blindness plays a major role in designing websites. Briefly explain a simple solution to make websites with colour-blind friendly. [20%]
- e) Briefly explain how a deaf person can use a computer system. [20%]



University of Vavuniya

Second Examination in Information Communication Technology - 2020

First Semester - April/May 2022

*Interconnective
Integration
Interactive*
TICT2142 - Multimedia Design and Technologies

Answer Four Questions Only

Time Allowed: Two hours

-
1. (a) Write four fundamental attributes of multimedia and briefly describe each of them. [20%]
 - (b) Draw the global structure of multimedia and illustrate the three domains associated with multimedia. [30%]
 - (c) State the difference between multimedia and hypermedia with suitable examples. [20%]
 - (d) Give five necessary elements to be included in the distribution of multimedia products on the Internet. *text, audio, video, Graphics, Animation* [05%]
 - (e) Analyze three major factors when considering the storage requirements for multimedia systems. [25%]

2. (a) State four major factors that are involved in choosing texts in multimedia applications and explain why these factors are important when creating multimedia applications. [25%]

(b) State the difference between font and font style. [15%]

(c) Define what is portable document format and what are other formats available in text. [20%]

(d) Define what is meant by sound and identify the factors that determine the sound quality. [20%]

(e) The size (S) of digital recording is represented as $S = R \times b \times C \times D$ bits, where R- sampling rate, C-Channel, b-resolution in bits, and D- recording duration (in seconds). A multimedia application is used to record a speech with a sampling rate of 11KHz, 8-bit resolution, and a single channel and stored on a hard disk. Find the recording duration in seconds if the hard disk has a free space of 512MB. [20%]

(a) Define what is digital image and identify the factors that affect the quality of a digital image. [20%]

(b) Briefly describe the benefits and drawbacks of having a higher resolution image. [15%]

(c) Define each of the following terms in the context of digital image.

i. Display resolution

ii. Pixel dimension

iii. Bit depth

iv. Grayscale image

(d) Explain the types of image data formats with aid of examples [20%]

(e) A bitmap image is represented with each of the resolutions of 320 x 240, 640 x 480, and 800 x 600 pixels for a multimedia application. Each pixel has 24 bits of depth. Find the size of the bitmap image in kilobytes in each of the resolutions. [25%]

communication system
Distributed
Tools & Application
Network, Storage

Interactive
Integration
* Compression

$$\begin{array}{r} 512 \\ 32 \\ \hline 544 \end{array}$$

$$\begin{array}{r} 96 \\ 96 \\ \hline 192 \end{array}$$

$$\begin{array}{r} 32 \\ 32 \\ 32 \\ \hline 96 \end{array}$$

$$\begin{array}{r} 32 \\ 15 \\ \hline 160 \\ 32 \\ \hline 480 \end{array}$$

$$\begin{array}{r} 480 \\ 32 \\ \hline 512 \end{array}$$

$$\begin{array}{r} 11760 \\ 56 \\ \hline 200 \\ 20 \\ \hline 220 \end{array}$$

$$\begin{array}{r} 11760 \\ 56 \\ \hline 200 \\ 20 \\ \hline 220 \end{array}$$

$$\begin{array}{r} 11760 \\ 56 \\ \hline 200 \\ 20 \\ \hline 220 \end{array}$$

$$\begin{array}{r} 11760 \\ 56 \\ \hline 200 \\ 20 \\ \hline 220 \end{array}$$

$$\begin{array}{r} 192 \\ 32 \\ \hline 224 \end{array}$$

4. (a) Give three examples of multimedia applications and explain how to deliver multimedia application. [25%]
- (b) What are the types of Multimedia Authoring Tools and explain them with aid of examples [25%]
- (c) Describe what is multimedia system and explain the types of multimedia system. [25%]
- (d) List the components of multimedia system and explain the multimedia system development cycle. [25%]
5. (a) Describe the major role of interactive multimedia in web designing and give three examples of interactive multimedia. [25%]
- (b) Differentiate the linear and non-linear interactive media with aid of an example. [20%]
- (c) State four challenges of multimedia system design and describe three desirable features of the multimedia system. [30%]
- (d) Give three important categories of multimedia authoring tools and describe each of them with aid of examples. [25%]



University of Vavuniya

Second Examination in Information Communication Technology - 2020

First Semester - March/April 2022

TICT2134 - Advanced Computer Programming (Theory)

Answer Four Questions Only

Time Allowed: Two hours

- ✓ 1. (a) Explain why the Main method is always static. [20%]
- (b) Write an example code snippet for the Main method to display a message Today is a Wonderful Day. [15%]
- ✗ (c) Describe the purpose of the 'Selection' statements in programming language. [20%]
- (d) Write the code snippet for each of the following keywords used in statements:
- i. default
 - ii. continue
 - iii. break [30%]
- (e) Describe the following statement. `System. Console.WriteLine (Happy);` [15%]
2. (a) Define the term arguments and provide an example of passing an argument in a method. [20%]
- (b) State the elements of the method declaration with an aid of an example. [20%]

[This question is continued on the next page]

(c) Differentiate the instance methods and static methods with an example for each of them. [20%]

(d) Compare and contrast the method param modifiers in, ref and out. [20%]

(e) Write the output for the following code snippet.

```
using System;
public class SamplePoint
{
    public int x;
    // Initialize a readonly field
    public readonly int y = 25;
    public readonly int z = 5;
    public SamplePoint()
    {
        // Initialize a readonly instance field
        z = 24;
    }
    public SamplePoint(int p1, int p2, int p3)
    {
        x = p1;
        y = p2;
        z = p3;
    }

    public static void Main()
    {
        SamplePoint p1 = new SamplePoint(11, 21, 32); // OK
        Console.WriteLine($"p1: x={p1.x}, y={p1.y}, z={p1.z}");
        SamplePoint p2 = new SamplePoint();
        p2.x = 55; // OK
        Console.WriteLine($"p2: x={p2.x}, y={p2.y}, z={p2.z}");
    }
}
```


3. ✓ (a) Explain the term Fully Qualified Name using a simple example. [10%]
- (b) State the main difference between the "Method Overloading" and "Method Overriding". [20%]
- (c) Discuss two types of arrays in C#? Give an example for each of the array types for declaration and initialization. [20%]
- (d) Write a program that performs the following tasks:
- i. A method to get the values from user to an integer typed array which is publicly declared.
 - ii. Search the maximum element within the above array. [20%]
- (e) Constructors are used to create an instance of a class.
- i. Define two types of Constructors with the aid of examples. [10%]
 - ii. Differentiate a method and a constructor with the aid of examples. [20%]
4. ✓ (a) Define the term "variables" with an example. [10%]
- (b) What is defined by the terms base class and derived class? How do you access a base class methods and constructors to a derived class. [20%]
- (c) Compare and contrast the **const** and **readonly** keywords. [20%]
- * (d) State the usage of namespace in C# with the aid of examples. [20%]
- (e) Compare and contrast the terms:
- i. **virtual** and **override**.
 - ii. **abstract** and **sealed**.
 - iii. **this** and **base**. [30%]

5. ✓ (a) Write a C# program to implement the following methods with proper delegates.

- i. To return the addition of two integers ToADD(int a, int b)
- ii. To return the square of a number ToSQUARE(int a)
- iii. To return the factorial of a number ToFACT(int a)

[40%]

(b) Explain the term Typecasting and its types by giving example for each of them. [25%]

(c) Implement the program skeleton for the given example and create an object for LibraryClass.

- i. checkPalin (string word)- returns 1 if the word is palindrome.
- ii. GetNthFibonacci_Rec (int n)- return the nth Fibonacci number in the Fibonacci series.

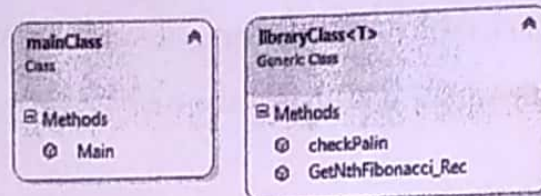


Figure 1: Skeleton of the program

[35%]

5x4x3x2x1
20
60
120



University of Vavuniya

Second Examination in Information Communication Technology - 2020

First Semester - March/April 2022

TICT2134 - Advanced Computer Programming (Practical)

Answer All Questions

Time Allowed: Three hours

1. A movie production company named "The Zee Studio" has planned to automate their Movie Details Management System and they have requested your help to accomplish it.

Consider the following class diagram in Figure 1.

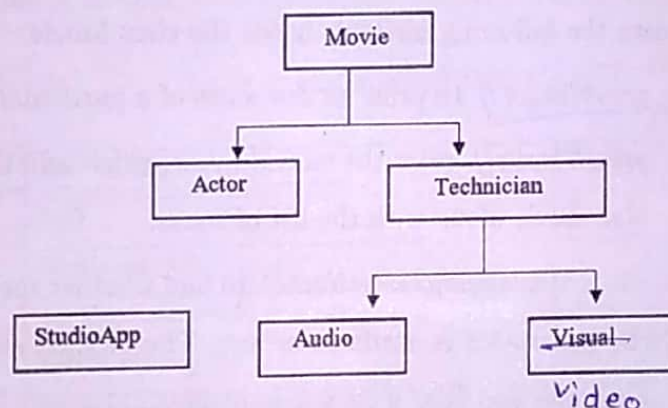


Figure 1: Class Diagram

The major task to map the above diagram into a computer program using C# programming language in order to full fill the request of "The Zee Studio".

Create all the classes inside the namespace **ZeeStudio** and program the C# application to satisfy the following requirements.

- (a) Create a parent class named Movie with 2 child classes Actors and Technicians. The Technician is having 2 child classes as Audio and Video. Movie: Class

✓ i. Create an instance variables mName, id, year, review, director, production-Company and an array songList and set the value of the variable prodComp as "Zee Pictures".

✓ ii. Create three constructors.

✓ A. A default constructor to initialize all instance variables as 0 or null.

✓ B. A parameterized constructor with the parameters mName and year.

C. Another parameterized constructor with the parameters mName, id, year, review and director.

The instance variables should be initialized inside the appropriate parameterized constructor using the values passed through the parameters.

✓ iii. Create separate properties to get and set the values of the instance variables; name, id, year, review, director, prodComp and songList inside the Movie class. Note that the prodComp property should be readonly.

iv. Create the following methods inside the class Movie.

(A. printSongs()) To print all the songs of a particular movie.

B. printDetails() print the name, id, year, review and the director of a particular movie along with the list of songs.

C. checkAvailability(movieName) to find whether the movie passed through the parameter is available or not. The method should return true if its available and false if its not available.

(b) Create a delegate named MyDelegate. The delegate should be of integer type with no parameters.

(c) Derive 2 classes Actor and Technician from the parent class Movie.

[15%]

Actor: Class

- (a) Class Actor should not be able to be extended by other classes.
- ✓(b) Declare the instance variables *salary*, *starRating*, *name*, *age* and *rank* and set the value of rank to 0.
- ✓(c) Create three constructors.
- A default constructor to initialize all instance variables as 0 or null.
 - A parameterized constructor with the parameters mName and year which calls the super class constructor.
 - Another constructor with the parameters salary, starRating, name and age to initialize these variables.
- (d) Create the following methods inside the class Actor.
- calculateRank()* To return the rank after calculating the rank based on the *starRating*. Consider the following table to calculate the rank. Use the delegate to execute this method.
 - Override the method *printDetails()* inside the class Actor to print the salary, starRating, name, age and rank of an actor. The method should also print the name of the production company from the parent class.

Table 1: Rank based Starrating

Star Rating	Rank
5.00	1
$\text{StarRating} \geq 4.5$	2
$4.0 \leq \text{StarRating} < 4.5$	3
$3.5 \leq \text{StarRating} < 4.0$	4
$\text{StarRating} < 3.5$	5
Else print message "Please check the Star Rating!"	

Technician: Class

- ✓(a) Create the instance variables *basicSalary* and *experience*.
- ✓(b) Create separate properties to get and set the values of the instance variables.
- ✓(c) Create the following methods inside the class Technician.
 - i. *calculateAllowance()* calculate and return the allowance. The allowance is calculated using the following equation.
$$\text{allowance} = (\text{basicSalary} * \text{experience}) / 10$$
 - ii. *calculateTotalSalary()* to calculate and return the total salary. Total salary is calculated using the following equation.
$$\text{totalSalary} = \text{basicSalary} + \text{allowance}$$

Audio: Class

- ✓(a) The class Audio should be an abstract class.
- ✓(b) Create an abstract method *rateAudioType ()* of type double.
- (c) Create a method *printSongType(duration)* to print a message Full Length Song if the duration is greater than 2.00 and a message BGM otherwise.

✓Video

Visual: Class

- (a) Create a variable named *type* and set it to null.

StudioApp: Class

- ✓(a) Create a class StudioApp which contains a main method.
- (b) Create the following movies inside the main method. Note that the Movies should be added using the object initializers.
- (c) Add the following songs to the movie Titanic.
$$\text{songs} = \text{"Never an Absolution", "Distant Memories", "Southampton", "Rose", "Leaving Port", "A Promise Kept", "A Life So Changed", "An Ocean of Memories", "My Heart Will Go On"}$$

Table 2: The object creation

Movie Name	ID	Year	Review	Director
Avengers: Endgame	001	2019	8.4	Anthony Russo
Titanic	002	1997	8.0	James Cameron

(d) Create necessary objects and methods to get the following output.

- Print all the movie names.
- Print the details of the second movie
- Get a movie name to check availability as a user input and print the availability.

The sample output is shown below.

Sample Output 1:

```

Welcome to Zee Studio App
-----
Movie Name: Titanic
Id: 2
Year: 1997
Review: 8
Director: James Cameron
-----
Songs List of the Movie:
1. Never an Absolution
2. Distant Memories
3. Southampton
4. Rose
5. Leaving Port
6. A Promise Kept
7. A Life So Changed
8. An Ocean of Memories
9. My Heart Will Go On
-----
Movie Names:
1. Avengers: Endgame
2. Titanic
-----
Enter the movie name to find:
Spiderman
Movie Not Available
-----
Please check the Star Rating!
Production Company: Zee Pictures
Salary: 150000
Star Rating: 4.5
Name: Robert Downey Jr.
Age: 56
Rank: 2
-----
Allowance: 20500
Total Salary: 41000
-----THANK YOU-----

```

Sample Output 2:

```

Welcome to Zee Studio App
-----
Movie Name: Titanic
Id: 2
Year: 1997
Review: 8
Director: James Cameron
-----
Songs List of the Movie:
1. Never an Absolution
2. Distant Memories
3. Southampton
4. Rose
5. Leaving Port
6. A Promise Kept
7. A Life So Changed
8. An Ocean of Memories
9. My Heart Will Go On
-----
Movie Names:
1. Avengers: Endgame
2. Titanic
-----
Enter the movie name to find:
Titanic
Movie Available!
-----
Please check the Star Rating!
Production Company: Zee Pictures
Salary: 150000
Star Rating: 4.5
Name: Robert Downey Jr.
Age: 56
Rank: 2
-----
Allowance: 20500
Total Salary: 41000
-----THANK YOU-----

```

Figure 2: Sample output



University of Vavuniya

Second Examination in Information Communication Technology - 2020

First Semester - May/June 2022

TICT2122 Statistics for Technology

Answer Four Questions Only

Time Allowed: Two hours

- ✓ 1. The masses of 50 apples (measured to the nearest g) were noted as in table.

86	101	114	118	87	92	93	116	105	102
97	93	101	111	96	117	100	106	118	101
107	96	101	102	104	92	99	107	98	105
113	100	103	108	92	109	95	100	103	110
113	99	106	116	101	105	86	88	108	92

- (a) Construct a stem and leaf diagram to represent these data. [10%]
- (b) Construct a frequency distribution table, using class widths of 5g, and taking the lower class of the first interval as 84.5g. [15%]
- (c) Construct a relative frequency histogram. [10%]
- (d) Find the following:
- i. Mean
 - ii. Mode
 - iii. Range

[This Question continues on next page.]

iv. Standard Deviation

v. Coefficient of Skewness

[25%]

(e) Find the five-number summary for the above data

[20%]

(f) Draw a box plot to represent these data.

[10%]

(g) Check whether there are any outliers on these data.

[10%]

2. The haemoglobin levels in a sample of 100 people were measured correct to one decimal place. The results are summarised in the table given below:

Haemoglobin level	Number of People
9.0 - 9.9	1
10.0 - 10.9	10
11.0 - 11.9	14
12.0 - 12.9	15
13.0 - 13.9	23
14.0 - 14.9	21
15.0 - 15.9	13
16.0 - 16.9	3

(a) Draw a frequency polygon graph for the above data.

[10%]

(b) Find the following:

i. Mean

ii. Mode

iii. variance

iv. Standard Deviation

[60%]

(c) Find the quartiles for the above data.

[30%]

3. (a) i. State the Addition Theorem of probability [10%]
- ii. A recent study shows that 15% of the used cars on the market have had their odometers set back. Furthermore, 5% of the used cars available have been involved in an accident. The study also shows that 18% have had their odometers set back or have been in an accident. Find the probability that a randomly selected used car has had its odometer set back and has been in an accident. [20%]
- (b) In a certain crime-ridden area, the probability that a house will be burgled on a particular night is 0.04. A burglar alarm is installed in a house in the area. If the house is burgled, the probability that the alarm will go off is 0.95. If the house isn't burgled, the probability that the alarm will go off by accident is 0.1. Find the probability that the house has been burgled, if the alarm has gone off. [20%]
- (c) i. State the Bayes' Theorem [10%]
- ii. A company has two plants to manufacture motorbikes. Plant I manufactures 80 percent of motor bikes, and plant II manufactures 20 percent. At Plant I 85% motorbikes are rated standard quality or better. At plant II only 65% motorbikes are rated standard quality or better. Suppose a standard quality or better bike is selected at random.
- A. what is the probability that it was manufactured by Plant 1? [20%]
- B. what is the probability that it was manufactured by Plant 2? [20%]
4. (a) What is probability distribution? [05%]
- (b) Define mean and standard deviation of a discrete random variable (X) of the probability distribution. [10%]
- (c) Construct a probability distribution for x; where x is the sum of 2 fair dice.
- i. Find the probability of rolling a sum of at least 6. [10%]
- ii. Find the probability of rolling a sum that is higher than 9. [10%]
- iii. Find the probability of rolling a sum that is even. [10%]

[This Question continues on next page.]

iv. Find the mean and standard deviation of the above probability distribution.

[10%]

- (d) i. Define the cumulative distribution function $F(x)$ of a discrete random variable.

[05%]

- ii. A fair six-sided die is rolled 8 times. Here is the probability distribution for the number of 3s.

x	P(x)
0	0.2326
1	0.3721
2	0.2605
3	0.1042
4	0.0260
5	0.0042
6	0.0004

A. Find the probability that between 3 and 5 rolls, inclusive, are 3s.

[10%]

B. Find the probability of at least two 3s.

[10%]

C. Find the cumulative distribution function of x.

[20%]

5. (a) Distinguish the following probability distribution:

i. Binomial Distribution

ii. Poisson Distribution

[20%]

- (b) Find the mean and variance of the above distributions.

[20%]

- (c) Assuming that one in 80 births in a case of twins, calculate the probability of 2 or more sets of twins on a day when 30 births occurs. Compare the results obtained by using

i. Binomial distribution

ii. Poisson distribution

[40%]

[This Question continues on next page.]

- (d) An insurance company has discovered that only about 0.1 percent of the population is involved in a certain type of accident each year. If its 10,000 policy holders were randomly selected from the population, what is the probability that not more than 5 of its clients are involved in such an accident next year?
($e \approx 2.71828$ and $e^{-10} = 0.000045$)

[20%]